



Fernando Peres Valverde

Industrial Digital Systems Engineer | Automation | Python | IIoT

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Driving license B

My [Portfolio](#)

AVAILABLE TO MOVE

Professional Summary

Industrial professional with 20+ years of experience in automation systems, electronics and software. My work sits at the intersection of factory-floor automation and modern data infrastructure using Python.

PLC + Electronics + Industrial Networks + Python + IIoT + Machine Learning applied to predictive maintenance.

Education

UNIVERSITY

Faculdade de Tecnologia de São Bernardo do Campo
FATEC – São Paulo, Brazil
IT for Business (2012 – 2015)

UNIVERSITY

Centro Universitário Fundação Santo André – São Paulo Brasil
Bachelor's degree in History (2005 – 2009)

TECHNICIAN

Technical course in telecommunications – State Technical School of São Paulo (2002 – 2003)

TECHNICIAN

Technical course in Electronics – State Technical School of São Paulo (2000 – 2001)

TECHNICIAN

Auto mechanics – SENAI São Paulo – Brazil (1997 – 1998)

Employment History

Industrial Automation & IIoT Engineer – Electro MF (2024–Present)

PLC programming, IIoT predictive maintenance systems (MotorWatch IQ), OPC UA/MQTT data pipelines, industrial retrofits, diagnostics of motors and drives.

Python developer & Digital

Solutions Consultant (2019 - 2025)

Automation tools, APIs, RPA, web systems, data processing, desktop applications.

Founding Partner – Sustine Technologies (2014 – 2019)

IT infrastructure and industrial software projects.

Computer and Electronics Technician (2002–2014)

Industrial electronics repair and diagnostics at component level.

Electronics Technician (2001 – 2002)

Installation and maintenance of alarm and electronic security systems

Truck Mechanic (1997 – 2000)

Truck repair

Core Skills

- Industrial automation — Siemens PLCs, HMIs, VFDs
- IIoT & Industry 4.0 — OPC UA (IEC 62541), MQTT, InfluxDB, Grafana, edge data pipelines
- Python development — PySide6 desktop applications, Django, Flask, FastAPI, Pandas, RPA, Scrapy, Selenium
- Electronics — component-level board repair and development

COURSES

- IBM DataScience Professional Certificate
- TIA Portal Programacion 2 – Siemens
- Mantenimiento Electronico – Toshiba
- Python: RPA, Django, Analisis de Datos
- Agile Methodologies, SCRUM
- Oracle PL-SQL

KEY TECHNOLOGIES

Siemens TIA Portal
S7-1200 / S7-1500
WinCC Unified
OPC UA
MQTT
InfluxDB
Grafana
Python
FastAPI
PySide6
Pandas
scikit-learn
IFM IO-Link
PROFINET
Modbus TCP
Industry 4.0
Predictive Maintenance

Featured Project

[MotorWatch IQ](#)

MotorWatch IQ – Industrial IoT predictive maintenance platform using Siemens PLC, IO-Link sensors, OPC UA, MQTT, InfluxDB, Grafana and Python ML for anomaly detection.

Designed and developed a complete Industrial IoT platform for condition monitoring and predictive maintenance of electric motors.

Architecture:

- Siemens S7-1200 PLC
- IFM IO-Link vibration sensors
- OPC UA → MQTT → InfluxDB pipeline
- Grafana dashboards
- Python analytics and machine learning

Features:

- Real-time monitoring
- Anomaly detection
- Fault classification
- Predictive maintenance
- Automated reporting

Full-stack industrial solution developed and integrated by a single engineer.

VOLUNTEER

La Zubia Volunteer Association

- Repair of electronic equipment at the Solidarity Store
- Digital learning for seniors: teaching portable technologies.

LANGUAGES

- Spanish fluent
- English fluent
- Portuguese native